QUIZ: Electron Configuration **Bartonian ID _____**

Write neatly ◦ Boxes are reserved for *final answers only* ◦ *Bartonian ID* must be written on each page ◦ Do not staple.

1) What is the ground state electron configuration of sodium (long form)?

1

2) What is the ground state electron configuration of phosphorus (long form)?

2

3) What is the ground state electron configuration of titanium (long form)?

3

4) What is the ground state electron configuration of antimony (long form)?

4

5) What is the ground state electron configuration of radium (long form)?

5

6) What is the ground state electron configuration of samarium (long form)?

6

7) What is the ground state electron configuration of beryllium (short form / noble gas shortcut)?

7

8) What is the ground state electron configuration of argon (short form / noble gas shortcut)?

8

9) What is the ground state electron configuration of zinc (short form / noble gas shortcut)?

9

10) What is the ground state electron configuration of silver (short form / noble gas shortcut)?

10

11) What is the ground state electron configuration of polonium (short form / noble gas shortcut)?

11

12) What is the ground state electron configuration of californium (short form / noble gas shortcut)?

12

For questions 13 – 18, answer "yes" or "no".

13) Is Rb^{+} isoelectronic with Kr?

13

14) Is O^{2-} isoelectronic with Na^{+}?

14

15) Is Mg isoelectronic with Na?

15

16) Is Te^{2-} isoelectronic with Xe?

16

17) Is Be isoelectronic with Sr?

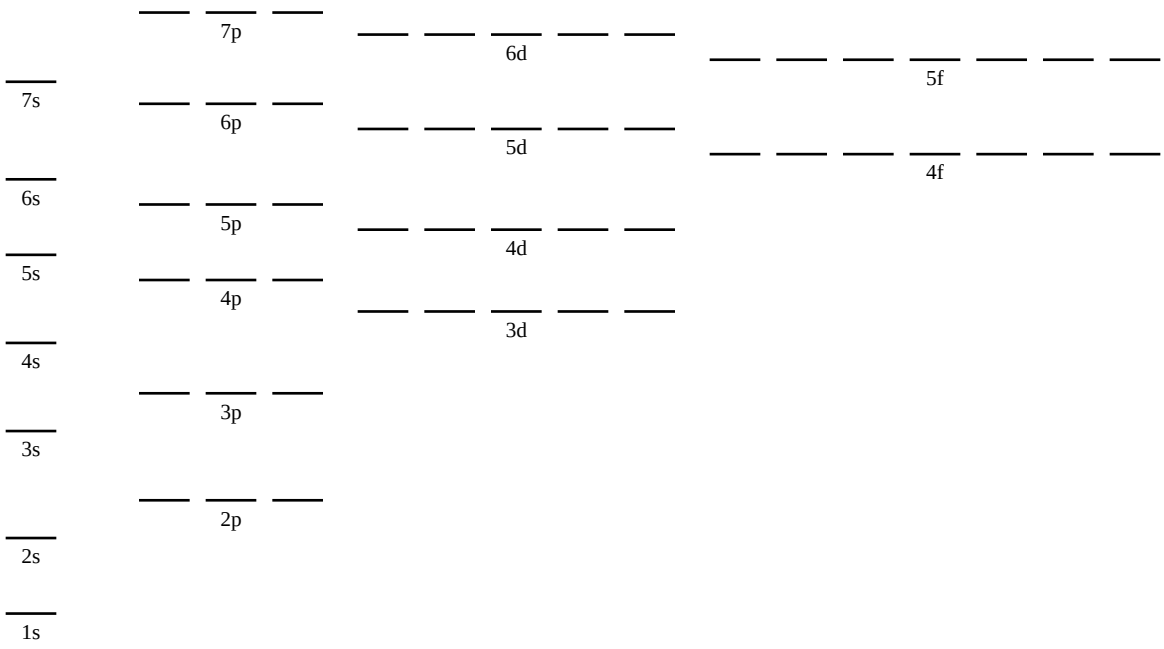
17

18) Is Sc^{3+} isoelectronic with Ar?

18

19) Draw an orbital diagram for boron in its ground state.

19



20) Draw an orbital diagram for nickel in its ground state.

20

